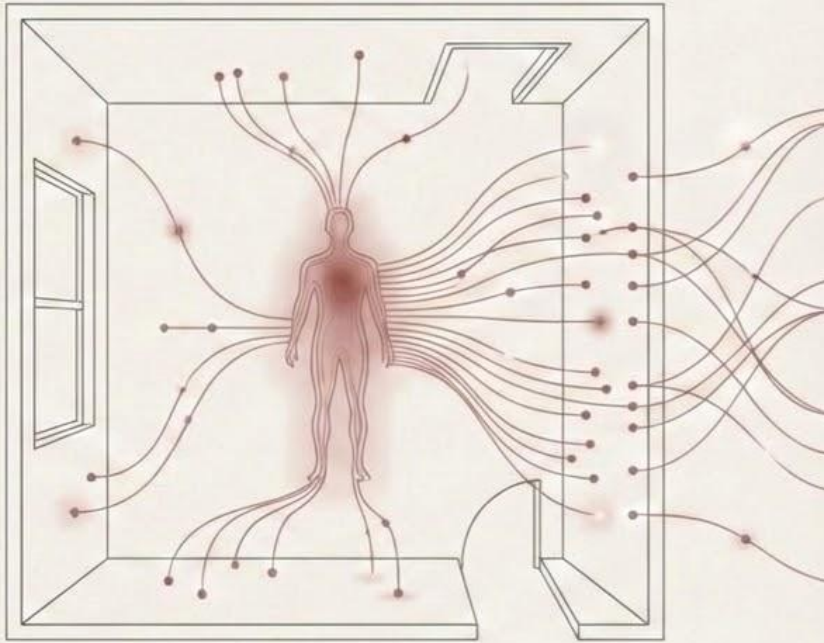
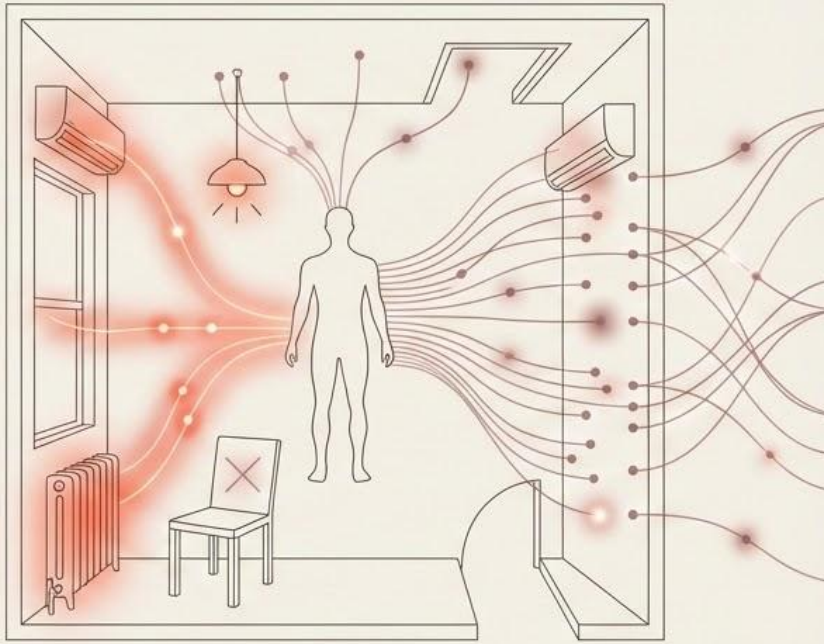
Smart Room Occupancy Monitoring from Sensor Streams

By Axel and Kyle



Problem Description

- Energy inefficiency is caused by heating, cooling, and lighting systems operating in unoccupied spaces.
- This project proposes a smart monitoring system that uses sensor data to detect room usage patterns in real time.
- By identifying when a room is occupied or unoccupied, the system can support more efficient energy management and automation.

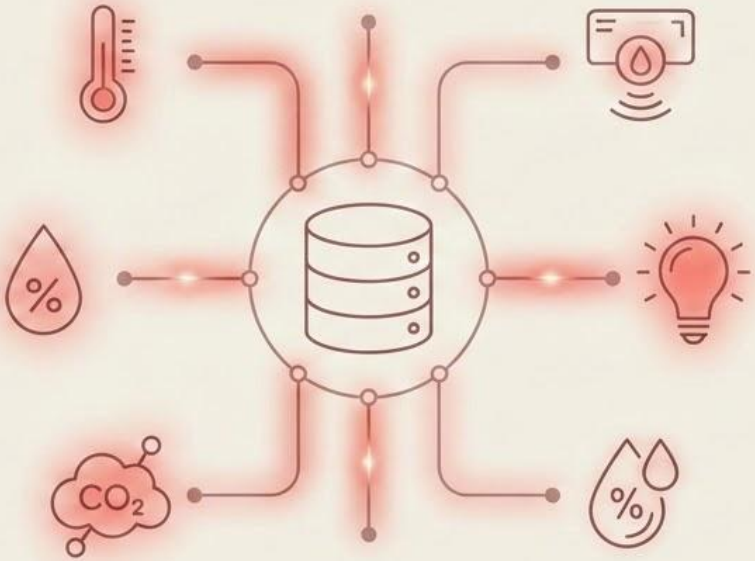


Unoccupied Inefficiency

Smart Monitoring

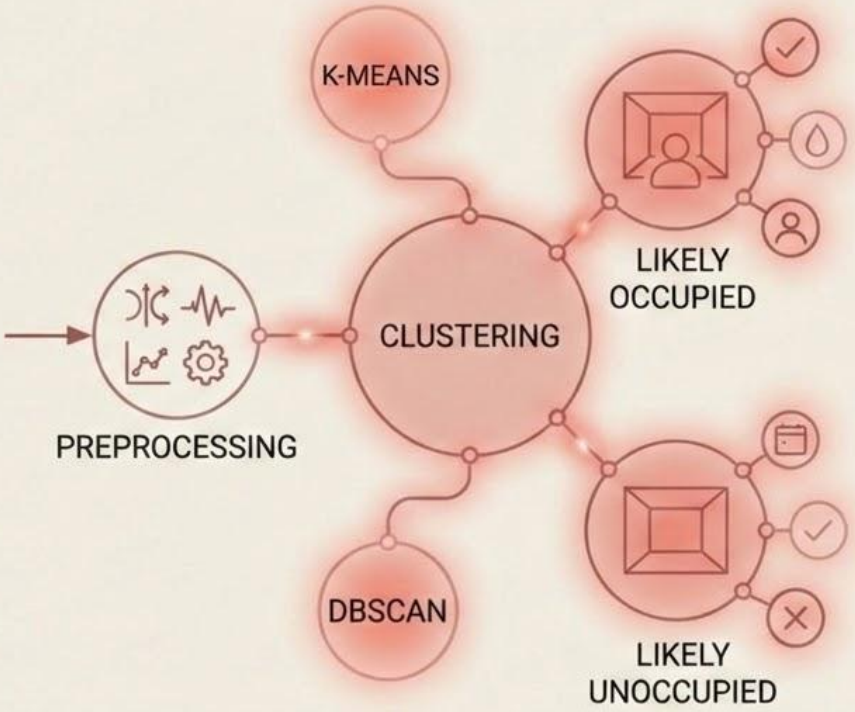
Dataset

- UCI Occupancy Detection Dataset
- 20,560 time-stamped records
- Features include temperature, humidity, light, CO2 levels, and humidity ratio
- Offers environmental context to infer human presence without cameras



Algorithms

- Clustering
  - K-Means
  - DBSCAN
- The clusters will be interpreted as occupancy states
  - Likely occupied vs Likely unoccupied
- Preprocessing will include normalization of sensor values, handling of time-series structure, and feature engineering.



Evaluation and Expected Outcomes

Evaluation

- Compare clustering to ground-truth labels
- Use internal metrics (e.g., silhouette score)
- Post-hoc labeling for accuracy, precision, recall, F1-score

Expected Outcomes

- Lightweight occupancy detection system
- Flag unusual sensor readings
- Integrate with HVAC & lighting
- Privacy-preserving, cost-effective, scalable

Thank you